

MSA - IATF 16949 Clause Mapping

Every MSA requirement mapped to IATF 16949:2016 clauses

QUALITY HEAD AGENTS | AIAG MSA 4th Edition | IATF 16949:2016

Clause 7.1.5 - Monitoring and Measuring Resources

7.1.5.1:	The organization shall determine and provide the resources needed to ensure valid and reliable monitoring and measuring results.
Requirement:	Measurement devices shall be suitable for the type, range, and accuracy of measurements required.
MSA Link:	GRR study proves the gauge is suitable. Without GRR, you cannot demonstrate clause 7.1.5.1.
Audit Evidence:	GRR study reports, gauge calibration records, gauge register

Clause 7.1.5.1 - Measurement System Analysis

Direct Requirement:	Statistical studies shall be conducted to analyze the variation present in the results of each type of monitoring and measuring equipment system.
Action Required:	GRR study before releasing gauge for production use. Re-study after gauge repair, process change, or 4M event.
Customer Specific:	Ford: GRR < 10% for all CC. GM: GRR < 20% for variable, Kappa > 0.75 for attribute. Stellantis: Full bias + linearity + stability required.
Audit Finding Risk:	Not performing GRR before first production run = Major NC (direct violation of automotive supplement).

Clause 7.1.5.2 - Calibration Records

Record Requirement:	Calibration records shall include: gauge ID, calibration date, next due date, reference standard traceability, calibration result, action when out of tolerance.
Traceability:	Must trace to NABL/NPL or equivalent national standard. Certificate must state traceability.
Audit Risk:	Out-of-calibration gauge found in use = Immediate Major NC. Products measured by out-of-cal gauge must be assessed for risk.

Clause 8.5.1.1 - Control of Special Characteristics

CC Requirement:	Critical Characteristics must be controlled with methods that guarantee detection of non-conforming product.
MSA Link:	GRR > 30% on a CC characteristic means your measurement system cannot reliably detect non-conformance. This is a direct violation.
Requirement:	CC gauges must demonstrate GRR < 10% (Ford requirement) or < 30% minimum (AIAG). Sample inspection not acceptable for CC without validated gauge.

Clause 8.3.4.4 - PPAP MSA Requirement

PPAP Element 8:	MSA studies (GRR, Bias, Linearity, Stability) must be submitted for approval before SOP.
Submission:	Include GRR report for all measurement systems used in control plan. Attribute GRR for go/no-go gauges.
Customer Review:	Customer PPAP reviewer will check: study method, operator count, part count, % GRR result, NDC.